Cluster Robust Variance Estimation

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Introduction

- To conduct hypothesis tests we need the asymptotic distribution of our estimators
- Usal variance estimator is not consistent in case of clustered data
- Cluster robust estimator exist but is not actually robust
- When to use it?

Which is the problem?

- Inside a cluster the scores of 2 individuals are not independent
- Usual White Variance estimator is not consistent

$$\frac{1}{\sqrt{N}} \sum_{i=1}^{N} X_i u_i \xrightarrow{d} \mathcal{N}(0,\ \Omega_N)$$

$$\Omega_N = \underbrace{\frac{1}{N} \sum_{i=1}^{N} \mathrm{Var}(X_i u_i)}_{\text{diagonal terms}} + \underbrace{\frac{2}{N} \sum_{i=1}^{N} \sum_{j \neq i}^{N} \mathrm{Cov}(X_i u_i, X_j u_j)}_{\text{off-diagonal terms}}$$

Literature Review

- Cameron and Miller (2015); Wooldridge (2003):
  Clustering corrects OLS standard errors for within-cluster dependence; validity relies on **large-G asymptotics**.
- **Ibragimov and Müller (2010)**:
  With fixed or small G, inference based on cluster-level estimates and a t_{G-1} test remains valid under heteroskedasticity.
- **Abadie et al. (2023)**:
  Clustering is justified by **sampling or assignment design**, not residual correlation per se.
- **Key implication**:
  If neither sampling nor treatment is clustered, **do not cluster**; CRVE is often **conservative**, especially with heterogeneous effects.

Representation of Clustered Data

- Sample with N individuals grouped in G clusters: $X_{ig} = (X_{11}, \dots, X_{N_1 1},\ X_{12}, \dots, X_{N_2 2}, \dots, X_{N_G G})$
- For i, j in the same cluster g:

$$P(X_i > E[X] \mid X_j > E[X]) > P(X_i > E[X])$$

- Cluster g can be seen as a **subpopulation**:

$$X_g = \{X_i : i \in g\}$$

Clustered Correlation from Heterogeneous Effects

- True model:

$$Y_{ig} = X_{ig}\beta_g + u_i$$

- Estimated model:

$$Y_{ig} = X_{ig}\beta_{\mathsf{APE}} + e_{ig}$$

- Residual:

$$e_{ig} = X_{ig}(\beta_g - \beta_{\mathsf{APE}}) + u_i$$

Heterogeneity-Induced Correlation

- Define cluster deviation:

$$d_g = \beta_g - \beta_{\mathsf{APE}}$$

- Between-cluster variance:

$$\sigma^2_{\mathsf{between}} = \sigma^2_{\mathsf{total}} - \sigma^2_{\mathsf{within}} = \mathrm{Var}(X_{ig} d_g)$$

- Hence, **heterogeneous** β_g **generates clustered residuals**.

Assumptions on the DGP

- **Two-stage sampling in all frameworks**:
  1. sample clusters
  2. sample individuals within clusters
- **Fixed number of clusters (G fixed)**
  - clusters are drawn once
  - repeated samples come only from re-sampling individuals
  - **only unit-level randomness**
- **Increasing number of clusters ($G \to \infty$)**
  - clusters and individuals are re-sampled each time
  - **two sources of randomness**:
    - cluster selection
    - unit-level sampling

Variance of Cluster-Level Effects

- Assume a population of cluster-specific effects:

$$\beta_g \sim \text{ i.i.d. with } E[\beta_g] = \beta_{\mathsf{APE}}, \quad \mathrm{Var}(\beta_g) = \sigma_\beta^2.$$

- We draw G coefficients:

$$\beta = (\beta_1, \dots, \beta_G)'.$$

- Then

$$E[\beta] = \begin{bmatrix} \beta_{\mathsf{APE}} \\ \vdots \\ \beta_{\mathsf{APE}} \end{bmatrix}, \quad \mathrm{Cov}(\beta) = \sigma_\beta^2 I_G.$$

Asymptotics in the Number of Clusters

- Sample mean of the cluster effects:

$$\bar{\beta} = \frac{1}{G} \sum_{g=1}^{G} \beta_g \xrightarrow{p} \beta_{\mathsf{APE}}$$

- Central Limit Theorem:

$$\frac{1}{\sqrt{G}} \sum_{g=1}^{G} (\beta_g - \beta_{\mathsf{APE}}) \xrightarrow{d} \mathcal{N}(0, \sigma_\beta^2).$$

- If all β_g were observed, we could estimate σ_β^2 via:

$$\hat{\sigma}_\beta^2 = \frac{G}{G-1} \sum_{g=1}^{G} (\beta_g - \bar{\beta})^2.$$

From True β_g to Estimated $\hat{\beta}_g$

- In practice, β_g is not observed; we estimate it from data.
- For each cluster g:

$$\sqrt{N_g}\,(\hat{\beta}_g - \beta_g) \xrightarrow{d} \mathcal{N}(0, V_g),$$

where

$$V_g = (X_g' X_g)^{-1} E_g[X_g' u_g u_g' X_g](X_g' X_g)^{-1}.$$

- Across clusters we thus have G asymptotically normal estimators,
  each centered at its own β_g with cluster-specific variance V_g.

Interpretation: Signal + Noise

- Write each estimator as

$$\hat{\beta}_g = \beta_g + e_g,$$

with

$$\sqrt{N_g}\, e_g \xrightarrow{d} \mathcal{N}(0, V_g), \quad E[e_g] = 0.$$

- The vector $\hat{\beta}$ can be viewed as **noisy observations** of the underlying cluster parameters, with noise coming from within-cluster sampling.

Decomposition Around β_{APE}

- Model the true effects as

$$\beta_g = \beta_{\text{APE}} + d_g, \quad d_g \sim \text{i.i.d. } \mathcal{N}(0, \sigma_\beta^2).$$

- Then

$$\hat{\beta}_g = \beta_{\text{APE}} + d_g + e_g.$$

- The cluster mean equals the pooled OLS estimator:

$$\hat{\beta}_{\text{pols}} = \frac{1}{G}\sum_{g=1}^{G}\hat{\beta}_g = \beta_{\text{APE}} + \frac{1}{G}\sum_{g=1}^{G} d_g + \frac{1}{G}\sum_{g=1}^{G} e_g.$$

- As $G \to \infty$, both averages of d_g and e_g converge to zero in probability.

White variance at the cluster level

- When clusters are fixed (**one-stage sampling**), the pooled White variance behaves like

$$\widehat{\mathrm{Var}}(\hat{\beta}) \approx \frac{1}{G^2} \sum_{g=1}^{G} \widehat{\mathrm{Var}}(\hat{\beta}_g),$$

  echoing the Fama–MacBeth logic of averaging group-level variances.
- Simulations show:
  - White consistently estimates the variance of each $\hat{\beta}_g$;
  - pooled White essentially aggregates these cluster-level variances;
  - but once clusters themselves are resampled, this equivalence breaks down and the pooled variance is no longer the average of the cluster-level ones.

Decomposition of $\hat{\beta}_g$

- Variation in $\hat{\beta}_g$ splits into:
  - **sampling error**: $e_g = \hat{\beta}_g - \beta_g$
  - **heterogeneity**: $d_g = \beta_g - \beta_{\text{APE}}$
- Simulations show:
  - $\mathbb{E}[\mathrm{Var}(e_g)]$ decreases quickly as N_g grows $\to$ large clusters yield precise $\hat{\beta}_g$.
  - $\mathrm{Var}(d_g)$ reflects true structural differences $\to$ does **not** shrink with N_g.
- Thus: sampling noise vanishes with large clusters, heterogeneity does not.

Behavior Across Cluster Sizes and Numbers

- Varying N_g: within-cluster error e_g declines, confirming improved precision.
- Varying G:

  $$\mathrm{Var}(\bar{d}_g)$$

  shrinks as more clusters are included $\rightarrow$ averaging many clusters recovers the mean effect reliably.
- Structural heterogeneity persists, but becomes easier to estimate as G grows.

Monte Carlo Decomposition of $\mathrm{Var}(\hat{\beta}_{\mathsf{ols}})$

- Goal: assess how well

  $$\mathrm{Var}(\hat{\beta}_{\mathsf{ols}}) \approx \mathrm{Var}(d_g) + \mathbb{E}\big[\mathrm{Var}(e_g)\big]$$

  holds in finite samples.
- Two components:
  - $d_g = \beta_g - \beta_{\mathsf{APE}}$ (heterogeneity)
  - $e_g = \hat{\beta}_g - \beta_g$ (sampling error)

Simulation Design

- Grid of designs:
  - cluster sizes $N_g \in \{20, 50, 100, 200, 300\}$
  - numbers of clusters $G \in \{5, 10, 20, 50, 100\}$.
- For each (N_g, G) and each repetition:
  1. Draw heterogeneous coefficients β_g.
  2. Generate a population with G clusters.
  3. Sample N_g individuals from each cluster.
  4. Estimate all $\hat{\beta}_g$ and pooled $\hat{\beta}_{\text{ols}}$.

Recorded Quantities

- For every repetition and design (N_g, G):
  - $e_g = \hat{\beta}_g - \beta_g$ (estimation errors)
  - $d_g = \beta_g - \beta_{\mathsf{APE}}$ (heterogeneity terms)
  - $\hat{\beta}_{\mathsf{ols}}$ (pooled estimator)
  - White and CRVE variance estimates of $\hat{\beta}_{\mathsf{ols}}$.
- For each (N_g, G) we compute:
  - $\mathrm{Var}(\hat{\beta}_{\mathsf{ols}})$ (empirical variance)
  - $\mathrm{Var}(d_g)$ and $\mathbb{E}[\mathrm{Var}(e_g)]$
  - relative discrepancy between left- and right-hand side.

Performance of the Decomposition

- When either G or N_g is large:
  - d_g and e_g are measured very precisely.
  - $\mathrm{Var}(d_g) + \mathbb{E}[\mathrm{Var}(e_g)]$

    closely matches $\mathrm{Var}(\hat{\beta}_{\text{ols}})$.
- When both G and N_g are small:
  - cluster-level estimates are noisy;
  - between-cluster dispersion is overstated;
  - the decomposition is a poorer approximation.

Relative Error Across Designs

- The plot of the relative error over (N_g, G) shows:
  - largest discrepancies when both G and N_g are small;
  - the mismatch rapidly shrinks as we increase either
    - the number of clusters, or
    - the cluster size.
- Conclusion: the variance decomposition is **very accurate** in rich designs (many clusters and/or large cluster sizes), but less so in sparse settings.

CRVE–White Differences as Heterogeneity Measure

- We test whether the **difference** between CRVE and White variance estimators
  captures the heterogeneity component

$$\mathrm{Var}(\bar{d}_g), \quad d_g = \beta_g - \beta_{\mathsf{APE}}.$$

- For each design (N_g, G):
  - compute average CRVE variance,
  - compute average White variance,
  - take their difference,
  - compare it to $\mathrm{Var}(\bar{d}_g)$ from the simulations.

Simulation-Based Comparison

- For every (N_g, G):
  - horizontal axis: $\mathrm{Var}(\bar{d}_g)$ (variance of mean deviations),
  - vertical axis:
    $$\mathbb{E}[\widehat{\mathrm{Var}}_{\mathsf{CRVE}}] - \mathbb{E}[\widehat{\mathrm{Var}}_{\mathsf{White}}].$$
- Each point is a design (N_g, G):
  - color intensity encodes G (lighter $=$ larger G).
- The 45-degree dashed line is the **benchmark** of perfect equality.

Results: Tight Alignment

- Points lie very close to the 45-degree line:
  - CRVE–White differences almost exactly recover $\mathrm{Var}(\bar{d}_g)$.
- Designs with **large** G (lighter points):
  - sit nearly on the identity line,
  - indicate very precise estimation of the heterogeneity component.
- Designs with **small** G:
  - show slightly more dispersion,
  - but still follow the same linear relationship.

Interpretation

- The gap between CRVE and White is driven by **between-cluster variation**, not by sampling noise.
- Overall, the simulations support:

$$\mathbb{E}[\widehat{\mathrm{Var}}_{\mathsf{CRVE}}] - \mathbb{E}[\widehat{\mathrm{Var}}_{\mathsf{White}}] \approx \mathrm{Var}(\bar{d}_g),$$

  and this relationship strengthens as the number of clusters G grows.

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